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Iris Flower Classification Project

Project Overview:

This project aims to build a predictive model for classifying iris flowers into three species: Setosa, Versicolor, and Virginica. The dataset contains botanical features such as sepal length, sepal width, petal length, and petal width. The objective is to classify each sample into one of the species using these features.

1. Importing Libraries:

We begin by importing the essential Python libraries for data analysis, visualization, and machine learning:

- `numpy` and `pandas` for numerical operations and data manipulation
- `matplotlib.pyplot` and `seaborn` for data visualization
- `sklearn` for preprocessing, model building, evaluation, and train-test splitting

2. Load Dataset:

The dataset (e.g., `iris.csv`) is loaded using `pandas.read_csv()`. Basic inspection using `.head()`, `.info()`, and `.describe()` helps understand data types, structure, and completeness.

3. Data Cleaning:

To ensure clean input for models:

- Missing values are checked and handled accordingly (if any)
- Duplicate rows are dropped to eliminate bias

4. Categorical Data Handling:

The target variable (species name) is a categorical column. It is encoded using `LabelEncoder` from `sklearn.preprocessing` to convert it into numerical values suitable for model training.

5. Feature Scaling:

All numerical features (sepal and petal dimensions) are scaled using `StandardScaler`, standardizing them to a mean of 0 and standard deviation of 1. This is crucial for algorithms like SVM and Logistic Regression.

6. Train-Test Split:

The dataset is split into:

- Training Set (80%) – used to train machine learning models
 - Testing Set (20%) – used to evaluate performance on unseen data
- `train_test_split()` from `sklearn.model_selection` is used for this.

7. Exploratory Data Analysis (EDA):

Visualizations help us understand the distribution and relationships:

- A countplot for each species
- A heatmap to show feature correlation
- Pairplots to visualize class separability across features

8. Feature Selection:

Feature relationships with the target are analyzed through visual plots and correlation matrices. All features in this dataset are informative and retained.

9. Model Training & Evaluation:

Four classification models are trained and evaluated:

Logistic Regression:

- Regularized linear classifier
- Fast and effective for multiclass classification

Decision Tree Classifier:

- Tree-based model with controlled depth to reduce overfitting

Random Forest Classifier:

- Ensemble of decision trees, better generalization and robustness

Support Vector Machine (SVM):

- Works well with scaled data and high-dimensional feature space

For each model:

- Trained on the training set
- Predictions are made on the test set
- Evaluation metrics include:
 - **Accuracy** – Proportion of correct predictions
 - **Precision** – Positive predictive value per class
 - **Recall** – Sensitivity for each class
 - **F1 Score** – Balance between precision and recall
 - **Confusion Matrix** – Summarizes prediction outcomes

10. Model Comparison:

All models are compared using accuracy and F1-score to assess generalization and precision. Based on metrics and interpretability, the most suitable model is selected for deployment.

Conclusion:

This project successfully implements a machine learning pipeline to classify iris species based on flower dimensions. It uses multiple algorithms and evaluation metrics to identify the best model. This foundation can be further extended to include:

- Hyper parameter tuning
- Cross-validation
- Deployment with Streamlit